Mock Exam: First Partial Exam

9 March 2023

1 Question 1

The economy of a country in which only the goods market exists is described by the following equations:

$$C = c_0 - c_1 T$$
$$I = \bar{I}$$
$$G = \bar{G}$$
$$T = \bar{T},$$

where $c_0 > 0$, $0 < c_1 < 1$ and the bar above a variable indicates that the variable is fixed and exogenous.

(a) Derive the expression of the equilibrium output for this economy. What is the expression of the multiplier?

(b) Assuming that in this economy we want to increase public spending while leaving the budget deficit unchanged, $\Delta\bar{G} = \Delta\bar{T} > 0$, how will the equilibrium output vary? What will be the consumption, investment, public and private savings in the new equilibrium? Why? Explain.

2 Question 2

True or False? Explain whether the following statements are true or false. Motivate your answer in a brief but rigorous way, making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

(a) An app-based payment technology reduces the fraction of money people hold in currency and, therefore, the M^s money supply shrinks.

(b) In the case inflation expectations are anchored ($\pi^e = \bar{\pi}$), the higher the parameter α is in the Phillips curve the greater deflation will be.

3 Question 3

Consider an economy described by an IS-LM-PC model based on the usual assumptions. In particular, taxes and public purchases are exogenous ($T = \bar{T}, G = \bar{G}$); the central bank conducts monetary policy by choosing the real policy rate, r; and investment depends negatively on the interest rate. Finally, assume that the inflation rate expected by individuals is $\pi^e = \pi_{t-1}$.

1. Graphically, represent the equilibrium in the medium-run, denoting it with the number 1. Suppose now that savings preferences increase (c_0 decreases). In the graph, denote the following short-run equilibrium by $1'$. Is this equilibrium stable? What can the central bank do to reach a new medium-run equilibrium where Y and π are the same as in the equilibrium 1? Denote the new medium-run equilibrium with 2. Compare the levels of consumption and investment in equilibrium before (equilibrium 1) and after (equilibrium 2).

2. Describe in detail how the answer to the previous question would change if the central bank conducted monetary policy by choosing the supply of money (exogenous money supply).

4 Question 4

Consider an economy that is initially in a medium-run equilibrium position and in which $\pi^e = \pi_{t-1}$. Assuming that the central bank chooses the interest rate, represent the initial medium-term equilibrium in the IS-LM-PC graphs and denote it by 1.

1. Suppose that the economy experiences a positive technological change, so that the parameter A of the production function $Y = AN$ becomes $A > 1$. In the graph, denote by 2 the new short-run equilibrium point that the economy will attain. How will the equilibrium level of production change in point 2 with respect to 1? What about the inflation rate? Why? Will it be necessary to adopt some monetary policy measures to arrive at a medium-run equilibrium?

2. Suppose now that, in equilibrium 1, the economy was in the *zero lower bound* (so that $i = 0$). How will your answer to the previous question change? In particular, is the short-run equilibrium in point 2 stable? Explain in detail. What other measures - other than that of the previous sub-question (4.1) - could be implemented to reach a medium-run equilibrium? Name two.

Mock Exam: First Partial Exam

9 March 2023

1 Question 1

The economy of a country in which only the goods market exists is described by the following equations:

$$\begin{aligned} C &= c_0 - c_1 T \\ I &= \bar{I} \\ G &= \bar{G} \\ T &= \bar{T}, \end{aligned}$$

where $c_0 > 0$, $0 < c_1 < 1$ and the bar above a variable indicates that the variable is fixed and exogenous.

(a) Derive the expression of the equilibrium output for this economy. What is the expression of the multiplier?

Answer: In equilibrium, production Y equals aggregate demand, Z, where

$$Z = C + I + G. \tag{1.1}$$

Substituting each component by their corresponding equations, we get to the expression of the equilibrium value of income:

$$Y^* = c_0 - c_1\bar{T} + \bar{I} + \bar{G}. \tag{1.2}$$

In equation (1.2), the right-hand side is a number called autonomous demand. The multiplier is, hence, equal to 1.

(b) Assuming that, in this economy, we want to increase public spending while leaving the budget deficit unchanged, $\Delta\bar{G} = \Delta\bar{T} > 0$, how will the equilibrium output vary? What will be the consumption, investment, public and private savings in the new equilibrium? Why? Explain.

Answer: After an identical increase in $\bar{G}$ and $\bar{T}$, equation(1.2) will change as follows:

$$\Delta Y^* = -c_1\Delta\bar{T} + \Delta\bar{G} = (1 - c_1)\Delta\bar{G} > 0, \tag{1.3}$$

where the second expression uses the fact that $\Delta\bar{G} = \Delta\bar{T}$ and the inequality is derived from the fact that $\Delta\bar{G} > 0$. Aggregate demand increases due to the increase in public spending and decreases due to the taxes. However, the positive effect of an increase in $\bar{G}$ is higher than the negative one of an increase in taxes because only part of the increase in $\bar{T}$ translates into a reduction in consumption (the fraction is c_1). As can be seen in Figure 1, in aggregate terms, demand will increase.

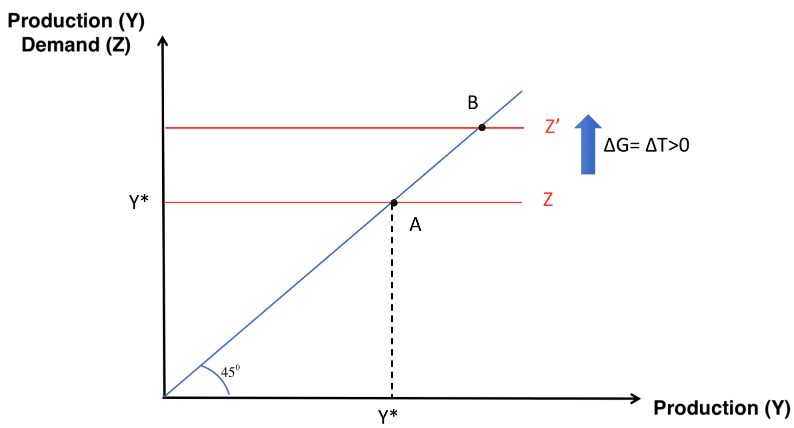

Figure 1: Change in equilibrium from A to B after an equal increase in G and T.

Regarding the rest of the variables, investment is given and exogenous, hence, it does not change. Consumption will decrease due to a change in taxes. Analytically,

$$\Delta C^* = -c_1 \Delta\bar{G} < 0. \tag{1.4}$$

Public savings, equal to T-G, will be unchanged by assumption. Finally, private savings will remain the same due to experiencing variations that cancel each other as follows:

$$\Delta S^* = \Delta Y^* - \Delta T^* - \Delta C^* = (1 - c_1)\Delta\bar{G} - \Delta\bar{G} + c_1\Delta\bar{G} = 0. \tag{1.5}$$

2 Question 2

True or False? Explain whether the following statements are true or false. Motivate your answer in a brief but rigorous way, making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

(a) An app-based payment technology reduces the fraction of money people hold in currency and, therefore, the M^s money supply shrinks.

False. *The new technology is equivalent to a reduction in parameter c. This will increase the amount of bank deposits. For a given parameter θ, an increase in bank deposits will allow banks to make more loans, which will in turn increase deposits further and again increase loans and deposits...and so the money supply will increase.*

Analytically, the money supply equation is:

$$M^s = \frac{\bar{H}}{c + \theta(1-c)}. \tag{2.1}$$

A decrease in the parameter c reduces the denominator and, therefore, M^s increases. The statement is false.

(b) In the case inflation expectations are anchored ($\pi^e = \bar{\pi}$), the higher the parameter α is in the Phillips curve the greater deflation will be.

False. *The general equation of the Phillips curve is as follows:*

$$\pi_t - \pi_t^e = (m+z) - \alpha u_t \tag{2.2}$$

and, imposing the condition $\pi_t^e = \bar{\pi}$, we obtain:

$$\pi_t = \bar{\pi} + (m+z) - \alpha u_t. \tag{2.3}$$

The highest the sensibility of salaries to changes in unemployment (α) the lower nominal wages that workers are willing to accept for a given increase in the unemployment rate. Thus, the cost of production firms face (W) goes down and firms will set lower prices. For a given P_{t-1}, π_t will be lower. Notice, though, that a lower inflation does not necessarily imply deflation (negative inflation). That would be the case only if the initial inflation rate was equal to zero. The statement is false.

3 Question 3

Consider an economy described by an IS-LM-PC model based on the usual assumptions. In particular, taxes and public purchases are exogenous ($T = \bar{T}, G = \bar{G}$); the central bank conducts monetary policy by choosing the real policy rate, r; and investment depends negatively on the interest rate. Finally, assume that the inflation rate expected by individuals is $\pi^e = \pi_{t-1}$.

1. Graphically, represent the equilibrium in the medium-run, denoting it with the number

1. Suppose now that savings preferences increase (c_0 decreases). In the graph, denote the following short-run equilibrium by $1'$. Is this equilibrium stable? What can the central bank do to reach a new medium-run equilibrium where Y and π are the same as in the equilibrium 1? Denote the new medium-run equilibrium with 2. Compare the levels of consumption and investment in equilibrium before (equilibrium 1) and after (equilibrium 2).

Answer: Graphically, the medium-run equilibrium corresponds to the point where the variation of inflation equals zero (that is, the PC curve intersects the horizontal axis). At that point, the level of production equals the natural output and the real interest rate is the natural real interest rate. We denote it in graph 2 with number 1.

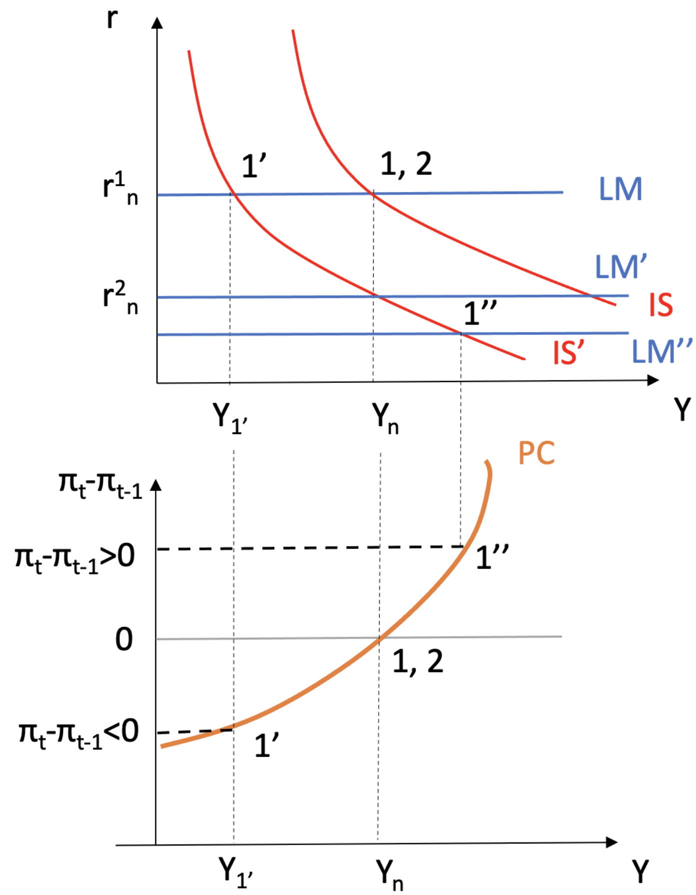

Figure 2: Short-run (point 1) and medium-run equilibrium (point 2) in a IS-LM-PC model, where $\pi_t^e = \pi_{t-1}$, after a shift in the IS curve to the left.

A decrease in c_0 brings about a reduction in aggregate consumption and, thus, aggregate demand and production in the equilibrium of the good's market. The IS curve moves to the left. The new short-run equilibrium, consequently, will be in point 1', where the new IS crosses the LM. At this point, inflation is decreasing (negative variation of the inflation rate). This might lead to deflation in the economy. Moreover, the short-term equilibrium production is below the potential production (negative output gap). All that makes equilibrium 1' unstable.

The central bank can decrease the policy rate and bring the economy back to the natural production level. The LM shifts downward until the LM' level in the graph. The new medium-run equilibrium is denoted by number 2. In the transition from equilibrium 1 to equilibrium 2, the successive negative variations in inflation accumulate over time. Hence, the resulting rate of inflation is lower in point 2 than it was in point 1. In order to reach the same level of inflation as in point 1, the central bank needs to reduce the policy rate r further, below r_n. This move makes the inflation level go up again. When inflation reaches the original level, r has to go back to the medium-run equilibrium r_n.

Comparing the values of consumption and investment in the medium-run equilibrium before and after, consumption is lower in equilibrium 2 because c_o has decreased by assumption, whereas investment is higher because the natural interest rate in 2 is lower than in 1. Moreover, given that the equilibrium value of production is the same in both equilibria, and $Y_n = C + I + \bar{G}$ where $\bar{G}$ is exogenous, the amount of the increase in investment is the same as the decrease in consumption.

2. Describe in detail how the answer to the previous question would change if the central bank conducted monetary policy by choosing the supply of money (exogenous money supply).

   Answer: The equilibrium value of Y in the short-run is lower when the central bank conducts monetary policy choosing the amount of money (exogenous money supply), as can be seen in graph 3. Therefore, the resulting output gap is lower and inflation decreases by less with respect to the case with endogenous money supply. In order to reach a new medium-run equilibrium, the central bank needs to increase the monetary base $\bar{M}$.

4 Question 4

Consider an economy that is initially in a medium-run equilibrium position and in which $\pi^e = \pi_{t-1}$. Assuming that the central bank chooses the interest rate, represent the initial medium-term equilibrium in the IS-LM-PC graphs and denote it by 1.

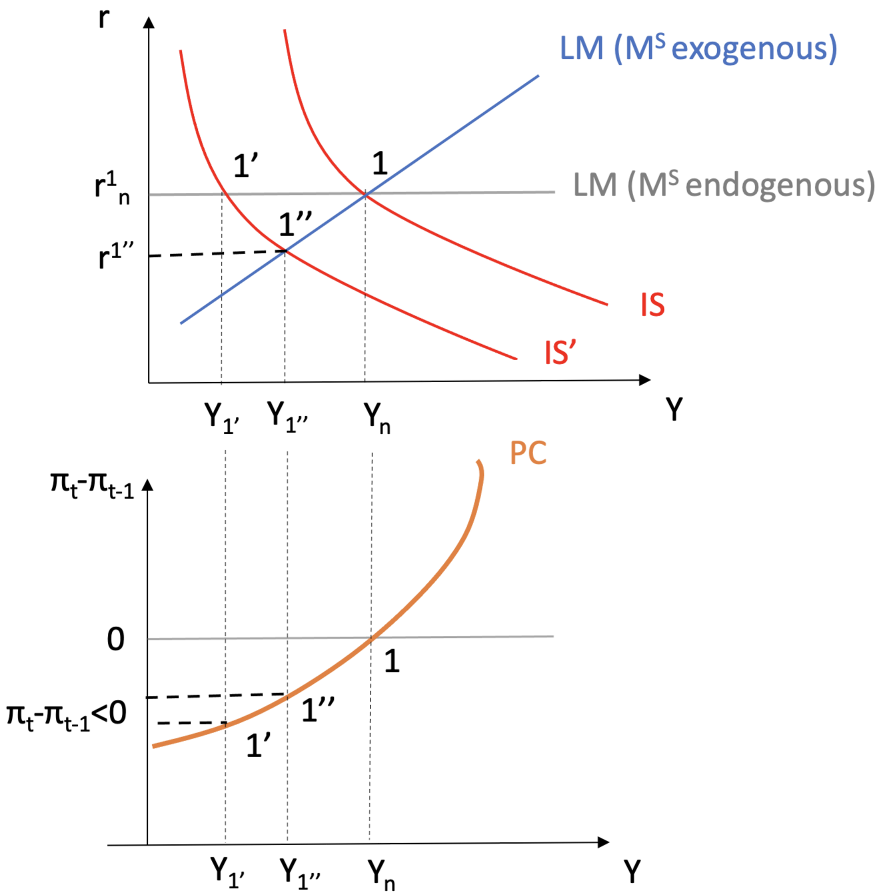

Figure 3: Short-run equilibrium after the IS curve shifts to the left under two different assumptions: endogenous money supply (point 1') and exogenous money supply (point 1").

1. Suppose that the economy experiences a positive technological change, so that the parameter A of the production function $Y = AN$ becomes $A > 1$. In the graph, denote by 2 the new short-run equilibrium point that the economy will attain. How will the equilibrium level of production change in point 2 with respect to 1? What about the inflation rate? Why? Will it be necessary to adopt some monetary policy measures to arrive at a medium-run equilibrium?

   Answer: In the labor market, represented graphically in figure 4, a positive technological change increases parameter A in the price-setting equation $\frac{W}{S} = \frac{A}{1+m}$. This shifts the PS upwards. In equilibrium, the natural unemployment rate decreases and, consequently, the natural production Y_n increases.

   In the IS-LM-PC graph in figure 5, the aforementioned change shifts the PC curve to the right. In the new short-run equilibrium (point 2 in the graph), income Y stays

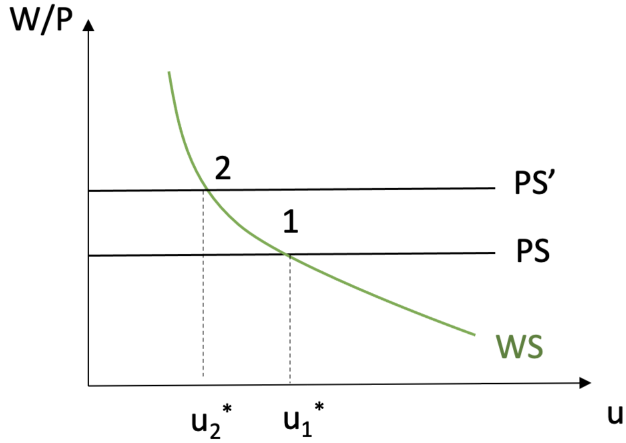

Figure 4: Labor market equilibrium after an increase in parameter A.

the same, because the equilibrium in the good's market and the financial market is unchanged, but the rate of inflation is decreasing, as can be seen in point 2 on the Phillips curve, due to the newly emerged negative output gap. In order to reach a new medium-run equilibrium (point 3), the central bank needs to intervene and bring the interest rate down. The LM line shifts downwards. In point 3, the output gap is zero, the interest rate is a lower natural interest rate and inflation is stable.

2. Suppose now that, in equilibrium 1, the economy was in the *zero lower bound* (so that $i = 0$). How will your answer to the previous question change? In particular, is the short-run equilibrium in point 2 stable? Explain in detail. What other measures - other than that of the previous sub-question (4.1) - could be implemented to reach a medium-run equilibrium? Name two.

   Answer: If the economy is in the zero-lower bound, that is $i = 0$, monetary policy becomes ineffective. Since $\pi_t - \pi_{t-1} < 0$ and $r = i - \pi$, where $i = 0$, it follows that the real interest rate r increases by the same amount that π decreases. Therefore, Y decreases, which brings π down and, again, makes r decrease and Y decrease etc. The economy is in a deflation spiral. We could undertake one of the following measures (or a combination of them) to get out of the deflation spiral: an expansionary fiscal policy (increase in G or decrease in T), that moves the IS to the right allowing the economy to reach the medium-run equilibrium, an increase in the worker's bargaining power (for instance, an increase in unemployment benefits) that makes z larger, or a reduction in the market power of firms (for instance, an antitrust legislation) that reduces m.

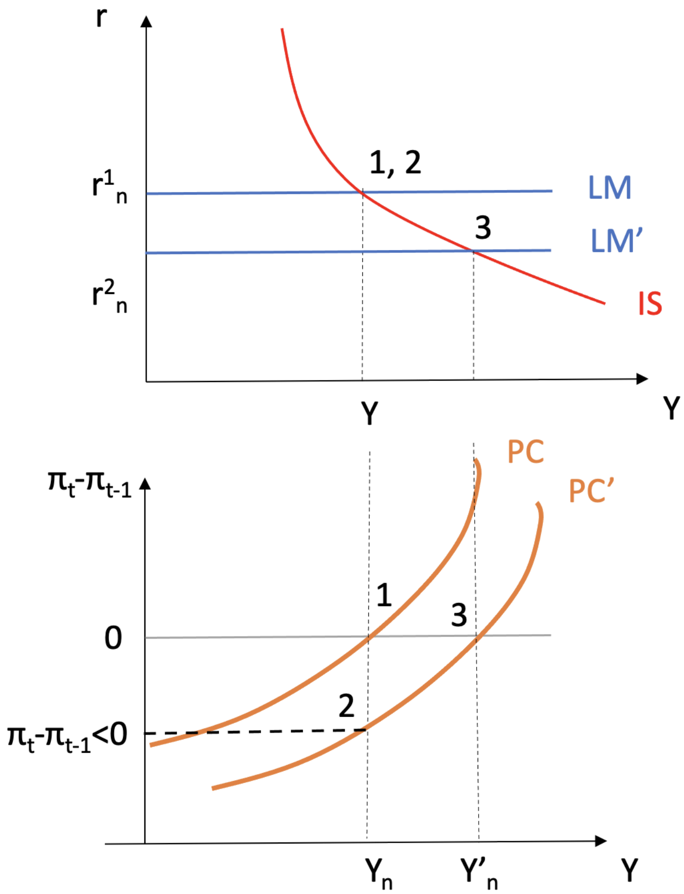

Figure 5: Short-run equilibrium (point 2) and medium-run equilibrium (point 3) after an increase in parameter A.